

REFERENCE SHEET 2024/2025

Question

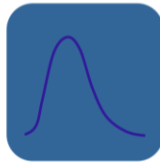
Acquire

Visualise

Model

Analyse

Conclude



Descriptive Statistics

Mean		Variance		Standard deviation	
Population $\mu = \frac{1}{N} \sum_{i=1}^N x_i$	Sample $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$	Population $\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$	Sample $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$	Population $\sigma = \sqrt{\sigma^2}$	Sample $s = \sqrt{s^2}$
Percentiles $[n(1-p)]$ values $\geq \eta(p)$		Median Odd n : $\tilde{x} = x_{\left(\frac{n+1}{2}\right)}$ Even n : $\tilde{x} = 0.5 \left(x_{\left(\frac{n}{2}\right)} + x_{\left(\frac{n}{2} + 1\right)} \right)$		Median absolute deviation $MAD = k \widetilde{x_i} - \tilde{x} $ $k = 1.4826$ if underlying distribution is normal	

Probability

Frequentist Probability $P(A) = n(A)/n$	Product rule $n = n_1 \times n_2 \times \dots \times n_k$	Permutations $P_{k,n} = \frac{n!}{(n-k)!}$	Combinations $C_{k,n} = \binom{n}{k} = \frac{n!}{k!(n-k)!}$
Intersection and Union $P(A \cup B) = P(A) + P(B) - P(A \cap B)$	Conditional probability $P(A B) = \frac{P(A \cap B)}{P(B)}$	Independence If $P(A B) = P(A)$ or $P(A \cap B) = P(A)P(B)$	Bayes Theorem $P(A_j B) = P(B A_j)P(A_j) / \sum_{i=1}^k P(B A_i)P(A_i)$

Random Variables

Expected Values			
Discrete $E(X) = \mu_X = \sum_x x p(x)$	Continuous $E(X) = \mu_X = \int_{-\infty}^{\infty} x f(x) dx$	Joint discrete $E[h(X,Y)] = \sum_x \sum_y h(x,y) p(x,y)$	Joint continuous $E[h(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x,y) f(x,y) dx dy$
Variance		Marginal PMF/PDF	
Discrete $V(X) = \sigma_X^2 = \sum_x (x - E(X))^2 p(x)$	Continuous $V(X) = \sigma_X^2 = \int_{-\infty}^{\infty} (x - E(X))^2 f(x) dx$	Discrete $p_X(x) = \sum_y p(x,y)$ $p_Y(y) = \sum_x p(x,y)$	Continuous $f_X(x) = \int_{-\infty}^{\infty} f(x,y) dy$ $f_Y(y) = \int_{-\infty}^{\infty} f(x,y) dx$
Independence		Covariance	
X and Y are independent if for every pair x, y		Discrete $cov(X,Y) = \sum_x \sum_y (x - E(X))(y - E(Y)) p(x,y)$	Continuous $cov(X,Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - E(X))(y - E(Y)) f(x,y) dx dy$
Discrete $p(x,y) = p_X(x)p_Y(y)$	Continuous $f(x,y) = f_X(x)f_Y(y)$		
Cumulative Distribution Functions		Other	
Discrete $F(x) = P(X \leq x) = \sum_{y: y \leq x} p(y)$	Continuous $F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy$	Percentiles $p = F[\eta(p)]$ $= \int_{-\infty}^{\eta(p)} f(y) dy$	Population Correlation coefficient $\rho = \frac{cov(X,Y)}{\sigma_X \sigma_Y}$

Key Distributions and Random Variables

Bernoulli $p(x; \alpha) = \begin{cases} 1 - \alpha & \text{if } x = 0 \\ \alpha & \text{if } x = 1 \end{cases}$ $E(X) = p, V(X) = pq$	Binomial $X \sim \text{Bin}(n, p)$ $b(x; n, p) = \binom{n}{x} p^x (1-p)^{n-x}$ $E(X) = np, V(X) = npq$	Poisson $X \sim \text{Pois}(\lambda)$ $\lambda = np, p(x; \lambda) = e^{-\lambda} \lambda^x / x!$ $E(X) = \lambda, V(X) = \lambda$
Normal $X \sim N(\mu, \sigma^2)$ $E(X) = \mu_X, V(X) = \sigma_X^2$	Standard Normal $Z = (X - \mu_X) / \sigma_X, Z \sim N(0, 1)$ $E(Z) = 0, V(Z) = 1$	Sample mean (CLT) $\bar{X} \sim N(\mu, \sigma^2/n)$ $E(\bar{X}) = \mu_X, V(\bar{X}) = \sigma_X^2/n$

Intervals

Confidence Interval		Prediction Interval	
$ME_{CI} = z_{1-\frac{\alpha}{2}}\sigma_{\bar{X}} = z_{1-\frac{\alpha}{2}}\sigma_X\sqrt{1/n}$ $ME_{CI} = t_{1-\alpha/2, \nu} s_X\sqrt{1/n}$	$CI = \bar{X} \pm ME_{CI}$ $[\bar{X} - ME_{CI}, \bar{X} + ME_{CI}]$	$ME_{PI} = z_{1-\frac{\alpha}{2}}\sigma_X\sqrt{1+1/n}$ $ME_{PI} = t_{1-\alpha/2, \nu} s_X\sqrt{1+1/n}$	$PI = \bar{X} \pm ME_{PI}$ $[\bar{X} - ME_{PI}, \bar{X} + ME_{PI}]$
For 2 sample analyses, replace $\bar{X}$, $s_{\bar{X}}$ and $ME_{\bar{X}}$ with corresponding $\bar{D}$ (paired) or $\bar{Y}$ (unpaired) values and DOF			

t and z tests

# of Samples	RV of interest	Standard error	Test Statistic	DOF
1 sample, x_i	$\bar{X}$	$\sigma_{\bar{X}} = \sigma_X/\sqrt{n}$	$Z = (\bar{X} - \mu_0)/\sigma_{\bar{X}}$	N/A
		$s_{\bar{X}} = s_X/\sqrt{n}$	$T = (\bar{X} - \mu_0)/s_{\bar{X}}$	$\nu = n - 1$
2 paired samples $d_i = x_{2,i} - x_{1,i}$	$\bar{D} = (\bar{X}_2 - \bar{X}_1)$	$\sigma_{\bar{D}} = \sigma_D/\sqrt{n}$	$Z = (\bar{D} - \mu_0)/\sigma_{\bar{D}}$	N/A
		$s_{\bar{D}} = s_D/\sqrt{n}$	$T = (\bar{D} - \mu_0)/s_{\bar{D}}$	$\nu = n - 1$
2 unpaired samples, x_1 and x_2	$\bar{Y} = \bar{X}_2 - \bar{X}_1$	$\sigma_{\bar{Y}} = \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$	$Z = (\bar{Y} - \mu_0)/\sigma_{\bar{Y}}$	N/A
		$s_{\bar{Y}} = \sqrt{s_1^2/n_1 + s_2^2/n_2}$	$T = (\bar{Y} - \mu_0)/s_{\bar{Y}}$	$\nu = \frac{(s_{\bar{X}_1}^2 + s_{\bar{X}_2}^2)^2}{s_{\bar{X}_1}^4/(n_1 - 1) + s_{\bar{X}_2}^4/(n_2 - 1)}$

Alternative hypothesis	Rejection region	P-value	Applies to $\mu = \mu_X$, $\mu = \mu_D$ & $\mu = \mu_Y$ $\Phi(z)$ is the CDF of the standard normal distribution $F(t, \nu)$ is the CDF of the t -distribution with ν DOF
$H_1: \mu > \mu_0$ upper tailed	$z \geq z_{1-\alpha}$	$p = 1 - \Phi(z)$	
	$t \geq t_{1-\alpha}$	$p = 1 - F(t, \nu)$	
$H_1: \mu < \mu_0$ lower tailed	$z \leq z_{\alpha}$	$p = \Phi(z)$	
	$t \leq t_{\alpha}$	$p = F(t, \nu)$	
$H_1: \mu \neq \mu_0$ two-tailed	$ z \geq z_{1-\alpha/2}$	$p = 2(1 - \Phi(z))$	
	$ t \geq t_{1-\alpha/2}$	$p = 2(1 - F(t , \nu))$	

Sample size and power (z-test)

Upper-tailed	$n = [\sigma(z_{1-\alpha} + z_{1-\beta})/(\mu_0 - \mu_1)]^2$	$\beta = \Phi\left(z_{1-\alpha} + \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}}\right)$
2-tailed	$n = [\sigma(z_{1-\alpha/2} + z_{1-\beta})/(\mu_0 - \mu_1)]^2$	$\beta = \Phi\left(z_{1-\alpha/2} - \frac{ \mu_0 - \mu_1 }{\sigma/\sqrt{n}}\right)$

Non-Parametric Tests

Sign Test (Asymmetric distribution)		
1 sample, x_i , or difference between 2 paired samples d_i		2 tailed $H_0: \tilde{\mu} = \tilde{\mu}_0$
Discard $x_i = \tilde{\mu}_0$ $s_i = \begin{cases} 0, & x_i - \tilde{\mu}_0 < 0 \\ 1, & x_i - \tilde{\mu}_0 > 0 \end{cases}$	Define binomial RV A: $a = \sum_{i=1}^n s_i$ $P(A = a) = b(a; n, 0.5)$	$H_1: \tilde{\mu} > \tilde{\mu}_0, p = P(A \geq a), H_1: \tilde{\mu} < \tilde{\mu}_0, p = P(A \leq a)$ $H_1: \tilde{\mu} \neq \tilde{\mu}_0: a < n/2, p = 2P(A \leq a),$ for $a > n/2, p = 2P(A \geq a)$
Sign-Rank Test (Symmetric distribution)		
1 sample, x_i , or difference between 2 paired samples d_i		2 tailed $H_0: \tilde{\mu} = \tilde{\mu}_0$
Discard $x_i = \tilde{\mu}_0$ $r_i = \text{sign}(x_i - \tilde{\mu}_0) \times \text{rank}(x_i - \tilde{\mu}_0)$	$w^+ = \sum_{i=1}^n r_i^+$ $w^- = -\sum_{i=1}^n r_i^-$ $w = \min\{w^-, w^+\}$	Rejection Region $H_1: \tilde{\mu} > \tilde{\mu}_0, w^- \leq w_{crit}$ $H_1: \tilde{\mu} < \tilde{\mu}_0, w^+ \leq w_{crit}$ $H_1: \tilde{\mu} \neq \tilde{\mu}_0, w \leq w_{crit}$
Rank-Sum Test		
2 unpaired samples, x_1 and x_2 ($n_1 \leq n_2$)		2 tailed $H_0: \tilde{\mu}_1 = \tilde{\mu}_2$
$r_i = \text{rank}\{x_1, x_2\}$ $w_1 = \sum_{i=1}^{n_1} r_{1,i}$ $w_2 = \sum_{i=1}^{n_2} r_{2,i}$	$u_i = w_i - \frac{n_i(n_i+1)}{2}$ $u = \min\{u_1, u_2\}$	Rejection Region $H_1: \tilde{\mu}_1 > \tilde{\mu}_2, u_2 \leq u_{crit}$ $H_1: \tilde{\mu}_1 < \tilde{\mu}_2, u_1 \leq u_{crit}$ $H_1: \tilde{\mu} \neq \tilde{\mu}_0, u \leq u_{crit}$
Kruskal-Wallis Test		
I unpaired samples (groups), $x_1, \dots, x_I$		$H_0: \tilde{\mu}_1 = \dots = \tilde{\mu}_I$
$r_i = \text{rank}\{x_1, \dots, x_I\}$ $w_1 = \sum_{i=1}^{n_1} r_{1,i} \dots$ $w_I = \sum_{i=1}^{n_I} r_{I,i}$	$N = \sum_{i=1}^I n_i$ $h = -3(N+1) + \frac{12}{N(N+1)} \sum_{i=1}^I \frac{w_i^2}{n_i}$	Rejection Region $H_1: \text{at least one of } \tilde{\mu}_i \text{ is different}$ $h > \chi_{\alpha}^2 \text{ with } \nu = I - 1$

Correlation and Linear Least Squares Regression

Sum Squares			
$S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 = SS_T$	$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$	$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$	$SS_R = \sum_{i=1}^n (y_i - \hat{y}_i)^2$
Sample correlation coefficients			
Pearson's $R(x, y) = S_{xy} / \sqrt{S_{xx} S_{yy}}$ $= \sqrt{1 - SS_R / SS_T}$	Spearman's $r_{x,i} = \text{rank}\{x\}, r_{y,i} = \text{rank}\{y\}$ $\rho_s = R(r_x, r_y)$	Test statistic $T = \frac{R\sqrt{n-2}}{\sqrt{1-R^2}}$	p-value p : as per t-tests but with R or $\rho = \mu$ and $\nu = n - 2$
Linear Regression			
Model: $Y = b_0 + b_1 X + \varepsilon$	Best fit: $Y = \hat{b}_0 + \hat{b}_1 X$	Residuals: $s_\varepsilon^2 = SS_R / \nu_R, \nu_R = n - 2$	
Slope		Intercept	
$\hat{b}_1 = S_{xy} / S_{xx}$	$s_{\hat{b}_1} = s_\varepsilon / \sqrt{S_{xx}}$	$\hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x}$	$s_{\hat{b}_0} = s_\varepsilon \sqrt{1/n + \bar{x}^2 / S_{xx}}$
$t_1 = \hat{b}_1 / s_{\hat{b}_1}$	$CI_1 = \hat{b}_1 \pm t_{1-\alpha/2, \nu_R} s_{\hat{b}_1}$	$t_0 = \hat{b}_0 / s_{\hat{b}_0}$	$CI_0 = \hat{b}_0 \pm t_{1-\alpha/2, \nu_R} s_{\hat{b}_0}$
Confidence and Prediction bands			
$s_{\hat{y}^*}(x^*) = s_\varepsilon \sqrt{1/n + (x^* - \bar{x})^2 / S_{xx}}$		$s_{pred}^2(x^*) = s_{\hat{y}^*}^2(x^*) + s_\varepsilon^2$	
$CI(x^*) = \hat{y}(x^*) \pm t_{1-\alpha/2, \nu_R} s_{\hat{y}^*}(x^*)$		$PI(x^*) = \hat{y}(x^*) \pm t_{1-\alpha/2, \nu_R} s_{pred}(x^*)$	

ANOVA

Grand Mean		Grand Variance	
$\bar{X} = \frac{1}{N} \sum_{k=1}^N X_k = \frac{1}{N} \sum_{i=1}^I \sum_{j=1}^{J_i} X_{ij}$		$s_{grand}^2 = \frac{1}{N-1} \sum_{k=1}^N (X_k - \bar{X})^2 = \frac{1}{N-1} \sum_{i=1}^I \sum_{j=1}^{J_i} (X_{ij} - \bar{X})^2$	
One-way (balanced/unbalanced)			
I groups	J_i observations/group	$N = \sum_{i=1}^I J_i$	$SS_T = SS_R + SS_M$
Total $SS_T = \sum_{i=1}^I \sum_{j=1}^{J_i} (X_{ij} - \bar{X})^2 = (N - 1) s_{grand}^2$ $\nu_T = N - 1$	Model $SS_M = \sum_{i=1}^I J_i (\bar{X}_i - \bar{X})^2$ $\nu_M = I - 1$	Residuals $SS_R = \sum_{i=1}^I \sum_{j=1}^{J_i} (X_{ij} - \bar{X}_i)^2 = \sum_{i=1}^I S_i^2 (J_i - 1)$ $\nu_R = N - I$	
One-Way Repeated Measures (balanced)			
I groups	J subjects	$N = IJ$	$SS_T = SS_B + SS_W, SS_W = SS_R + SS_M$
Total $SS_T = \sum_{i=1}^I \sum_{j=1}^J (X_{ij} - \bar{X})^2 = (N - 1) s_{grand}^2$ $\nu_T = N - 1$	Model $SS_M = \sum_{i=1}^I J (\bar{X}_i - \bar{X})^2$ $\nu_M = I - 1$	Residuals $SS_R = \sum_{i=1}^I \sum_{j=1}^J (X_{ij} - (\bar{X}_i + \bar{X}_j - \bar{X}))^2$ $\nu_R = \nu_W - \nu_M = N - J - (I - 1)$	
Between $SS_B = \sum_{j=1}^J I (\bar{X}_j - \bar{X})^2$ $\nu_B = J - 1$		Within $SS_W = \sum_{j=1}^J \sum_{i=1}^I (X_{i,j} - \bar{X}_j)^2 = (I - 1) \sum_{j=1}^J s_j^2$ $\nu_W = J(I - 1) = N - J$	
Two-Way (balanced)			
G groups for Factor A, I groups for Factor B	J observations/group	$N = GIJ$	$SS_T = SS_R + SS_M$ $SS_M = SS_A + SS_B + SS_{A \times B}$
Total $SS_T = \sum_{g=1}^G \sum_{i=1}^I \sum_{j=1}^J (X_{gij} - \bar{X})^2$ $\nu_T = N - 1$	Model $SS_M = \sum_{g=1}^G \sum_{i=1}^I J (\bar{X}_{gi} - \bar{X})^2$ $\nu_M = GI - 1$	Residuals $SS_R = \sum_{g=1}^G \sum_{i=1}^I \sum_{j=1}^J (X_{gij} - \bar{X}_{gi})^2$ $\nu_R = N - GI$	
Factor A $SS_A = IJ \sum_{g=1}^G (\bar{X}_g - \bar{X})^2$ $\nu_A = G - 1$	Factor B $SS_B = GJ \sum_{i=1}^I (\bar{X}_i - \bar{X})^2$ $\nu_B = I - 1$	Interactions $SS_{A \times B} = SS_M - SS_A - SS_B$ $\nu_{A \times B} = \nu_M - \nu_A - \nu_B$	

Mean squares, test statistics and p-values

One-Way	$MS_M = SS_M/\nu_M$	$MS_R = SS_R/\nu_R$	$F = MS_M/MS_R$	$p = 1 - f(F, \nu_M, \nu_R)$
2-Way, Factor A	$MS_A = SS_A/\nu_A$		$F_A = MS_A/MS_R$	$p_A = 1 - f(F_A, \nu_A, \nu_R)$
2-Way, Factor B	$MS_B = SS_B/\nu_B$		$F_B = MS_B/MS_R$	$p_B = 1 - f(F_B, \nu_B, \nu_R)$
2-Way, interaction	$MS_{A \times B} = SS_{A \times B}/\nu_{A \times B}$		$F_{A \times B} = MS_{A \times B}/MS_R$	$p_{A \times B} = 1 - f(F_{A \times B}, \nu_{A \times B}, \nu_R)$
Rejection region: H_1 : at least one of μ_i is different $f \geq f_{crit}$				